

WP-14 Literature Review — Residue Stress Sweep (Run 03)

The Privacy-is-Value Research Programme

2026-08-17

Lit-review runtime · run 03 manifest (residue stress sweep)

Purpose. Run 02 (wf_ba400906-67b) returned 0 VALIDATED / 5 MIRAGE and left five residual novel cores. Those residues are the claims WP-14 now rests on, so this run stresses the residues themselves against a widened, time-advanced corpus (2023-2026 window, searched 2026-08-17).

Seat configuration

Seat	Count	Role	Sees
sweep:*	5	multi-modal prior-art search (WEIS/econ venues; formal preprints; law+standards; data-as-labour; shelf-life/erosion)	its modality brief + the PVM position + the five residues + WebSearch
refute:*	5	per-residue DELEGATOR - hunts covering art, instructed to default to coverage	the residue + sweep digest + WebSearch; never a prover argument (D3)
judge	1	rules SURVIVES / NARROWED / COVERED per residue; assembles bibliography additions	residues + refuter outputs + sweep items

D4b honest limit (unchanged from run 02). All seats run on the same model weights; the separation is context isolation, not weight diversity. The operating rule stands: an all-SURVIVES result would be treated as a failed enforcement of the adversarial discipline. This run returned 1 COVERED / 4 NARROWED / 0 SURVIVES across 5 residues.

Run statistics

- Sweep items surfaced: 76 across 5 modalities

- Refuter covering candidates: 48
- Bibliography additions proposed: 74
- Verdict spread: **1 COVERED / 4 NARROWED / 0 SURVIVES across 5 residues**

Judge headline

1 COVERED, 4 NARROWED, 0 SURVIVES: LR-01's Phi/R assessment-unit move is anticipated outright (Nissim 2018, Abowd-Schmutte 2019, Census DSEP, NIST SP 800-226, Epsilon Registry) and should be retired into LR-02/LR-05; every other residue loses its per-leg firsts to named art and survives only as a joint composition — the priced adversary-relative drifting-capital quadruple (LR-02), the WTA-WTP/IEEE-7012 single-share-function identification (LR-03, strongest residue: no game-theoretic treatment of MyTerms exists anywhere), the Bajari-routed self-defeat falsifier landing on venue rent (LR-04, hardest-narrowed: Mohades-Savona 2026 already states the rent-not-wage conclusion by another route), and the Mosca-form reconstruction-clock substitution carried into subject-owned valuation (LR-05).

Outputs (this directory)

- `residue_verdicts_03.md` - per-residue verdict, restated residual, basis, and the refuter's covering art.
- `bibliography_additions_03.md` - new resolvable citations proposed for the WP-14 bibliography (A4 verifies before tier-A use).
- `_run_data_03.json` - full structured run output (sweeps, refutations, judge).

Residue verdicts: run-02 residues stress-tested (run 03)

Verdict spread: **1 COVERED / 4 NARROWED / 0 SURVIVES across 5 residues**. Vocabulary: SURVIVES = the residue stands as stated; NARROWED = partial coverage forces a smaller residue (re-stated below); COVERED = the residue as stated is anticipated. Confidence labels follow run-02 practice; percentages are runtime-internal and do not enter tier-A artefacts (GR-2).

CTR-LR-01 — COVERED (high, 82%)

Residual restatement (what now survives). As stated — a governance-facing, independently verifiable mechanism property Phi, held two-level-distinct from the per-subject bound R it gates, as the SINGLE UNIT of a privacy assessment/valuation instrument — the move is anticipated: Nissim et al. 2018 conduct a compliance assessment over the mechanism guarantee while holding a per-subject adversary-inference bound as the gated requirement; Abowd-Schmutte 2019 make epsilon the single unit of a social-choice valuation instrument; Census DSEP practice governs a per-subject RECONSTRUCTION threat by deliberating and setting epsilon; NIST SP 800-226 and the Epsilon Registry (Dwork-Kohli-Mulligan 2019, OpenDP) institutionalize the mechanism property as the assessed, comparable unit; the DP-auditing literature (arXiv:2210.08643 line) supplies the independent-verifiability qualifier; Kaissis 2023 / Biswas 2024-2025 supply the Phi-gates-R formal bridge. What survives is not an independent claim: only the instantiation where R is WP-14's time-drifting adversary-relative R(t) and the instrument is a subject-side capital VALUATION — that sliver folds into CTR-LR-02/CTR-LR-05 and should be merged there, with CTR-LR-01 retired as a separate novelty claim and rewritten as positioning/related-work.

Basis. Five strong candidates from independent institutional strands (legal formalization, economics, deployed census governance, standards, registry) jointly span the claimed composition, including reconstruction as the gated per-subject threat (Census 2020 DAS); the refuter itself recommended COVERED and its absence report concedes conjunct pairs co-occur in single works. The only uncovered triple is exactly the LR-02/LR-05 subject-matter, not the Phi/R assessment-unit move.

Covering art found by the refuter seat.

- [strong] Kobbi Nissim, Aaron Bembeneke, Alexandra Wood, Mark Bun, Marco Gaboardi, Urs Gasser, David R. O'Brien, Salil Vadhan, Thomas Steinke — Bridging the Gap between Computer Science and Legal Approaches to Privacy — Harvard Journal of Law & Technology 31(2) (2018) — A governance/compliance-assessment methodology whose two steps are exactly the two-level separation: (1) formalize the legal requirement as a per-subject adversary-inference bound (the R-level object), (2) prove the mechanism-level guarantee (DP, the Phi-level object) implies it. The assessment verdict (FERPA compliance) is conducted over Phi alone while the per-subject bound is held distinct as the gated requirement — a single work doing the two-level Phi/R separation AS the unit of a governance instrument. <https://www.semanticscholar.org/paper/Bridging-the-Gap-between-Computer-Science-and-Legal-Nissim-Bembeneke/22067befe017dc5760be667bf0ba9cf48b2ab650>
- [strong] John M. Abowd, Ian M. Schmutte — An Economic Analysis of Privacy Protection and Statistical Accuracy as Social Choices — American Economic Review 109(1); arXiv:1808.06303 (2019) — A valuation instrument (marginal-cost-equals-marginal-benefit social choice) whose single choice unit is epsilon, the mechanism property, with epsilon's inferential-disclosure semantics (bound on adversary posterior odds about an

individual) as the per-subject quantity it gates. Covers the ‘Phi as the single unit of a valuation instrument’ branch directly, in an economics venue with governance intent. <https://arxiv.org/abs/1808.06303>

- [strong] NIST — Guidelines for Evaluating Differential Privacy Guarantees — NIST SP 800-226 (final) (2025) — An institutional, governance-facing assessment instrument whose unit of evaluation is the mechanism guarantee itself (including how to interpret what it bounds for individuals and how implementations can be independently checked) — the closest standards-body embodiment of ‘assess Phi, not per-subject outcomes’. <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-226.pdf>
- [strong] U.S. Census Bureau (Abowd et al.) — The modernization of statistical disclosure limitation at the U.S. Census Bureau (2020 DAS: DSEP committee sets the privacy-loss budget in response to the reconstruction attack) (2020) — A deployed governance process in which a per-subject reconstruction threat (the 2010 census reconstruction attack — the R-level event) is governed by deliberating and setting the mechanism property epsilon (the Phi-level unit) via a policy committee (DSEP/DRB). The assessment/decision unit is Phi; reconstruction risk is what it gates. Institutional practice of the exact composition, minus the word ‘valuation’. <https://www.census.gov/content/dam/Census/library/working-papers/2020/adrm/The%20modernization%20of%20statistical%20disclosure%20limitation%20at%20the%20U.S.%20Census%20Bureau.pdf>
- [strong] Cynthia Dwork, Nitin Kohli, Deirdre Mulligan — Differential Privacy in Practice: Expose your Epsilons! — Journal of Privacy and Confidentiality 9(2); plus the live OpenDP Differential Privacy Deployments Registry and NIST IR registry-governance report (2025) (2019) — Positions the mechanism property epsilon as the public, comparable, oversight-enabling governance unit across deployments (the Epsilon Registry), explicitly for accountability pressure — Phi as the single governance-facing assessment unit, now institutionally realized at registry.opendp.org. <https://journalprivacyconfidentiality.org/index.php/jpc/article/view/689>
- [partial] Arpita Ghosh, Aaron Roth — Selling Privacy at Auction — EC 2011 / Games and Economic Behavior; arXiv:1011.1375 (2011) — A valuation/market instrument in which epsilon (the mechanism property) is literally the priced and procured unit, chosen from individuals’ privacy valuations — covering ‘Phi as the single unit of a valuation instrument’, though the gated per-subject quantity is generic privacy loss rather than a named reconstruction bound. <https://arxiv.org/abs/1011.1375>
- [partial] Justin Hsu, Marco Gaboardi, Andreas Haeberlen, Sanjeev Khanna, Arjun Narayan, Benjamin C. Pierce, Aaron Roth — Differential Privacy: An Economic Method for Choosing Epsilon — CSF 2014; arXiv:1402.3329 (2014) — An economic instrument that converts per-participant expected harm (an adversary-relative, per-subject quantity) into the choice of the mechanism parameter epsilon — the Phi/R direction of travel embedded in a valuation procedure, though it collapses the two levels into a calibration rather than holding Phi as an audited standing unit. <https://arxiv.org/abs/1402.3329>
- [partial] Georgios Kaissis, Jamie Hayes, Alexander Ziller, Daniel Rueckert — Bounding data reconstruction attacks with the hypothesis testing interpretation of differential privacy — arXiv:2307.03928 (with Riess et al. arXiv:2402.12861 and Balle-Cherubin-Hayes IEEE S&P 2022 as siblings) (2023) — Supplies the formal two-level structure the residue relies on: a mechanism-level DP property that provably gates a per-subject/per-record reconstruction bound. Not a governance or valuation instrument, but it removes any novelty in the Phi-gates-R separation itself. <https://arxiv.org/abs/2307.03928>
- [partial] Matthew Jagielski et al. lineage / (authors per arXiv) — A General Framework for

Auditing Differentially Private Machine Learning — NeurIPS 2022; arXiv:2210.08643 (2022) — The DP-auditing literature makes the mechanism property Φ empirically and independently verifiable — covering the ‘independently verifiable’ qualifier of the residue’s Φ . <https://arxiv.org/pdf/2210.08643>

- [partial] Ruwimal Y. Pathiraja, Jerome P. Reiter — Setting the Privacy Budget in Differential Privacy by Bounding Adversaries’ Odds of Learning Sensitive Information — arXiv:2607.04004 (2026) — Practitioner-facing procedure translating an adversary-posterior bound (R-level) into the epsilon setting (Φ -level) — a governance-usable two-step bridge, though it collapses the levels into a calibration rather than keeping Φ as the standing audited unit. <https://arxiv.org/abs/2607.04004>

Absence report (D2). `not_found_in_scope`(a single published instrument that simultaneously (i) names a time-drifting, adversary-relative per-subject RECONSTRUCTION bound $R(t)$ as the gated quantity, (ii) holds an independently auditable mechanism property Φ two-level-distinct from it as the standing assessed unit, and (iii) is framed as a privacy VALUATION/pricing instrument for data-as-capital. Searched: economics-of-epsilon literature (Abowd-Schmutte AER 2019, Hsu et al. CSF 2014, Ghosh-Roth EC 2011, federated data-trading pricing), epsilon-governance/registry literature (Dwork-Kohli-Mulligan 2019, OpenDP Deployments Registry, NIST IR 2025), standards/PIA instruments (NIST SP 800-226, CSRC PIA definitions, PIA tool guides), legal-compliance formalizations (Nissim et al. Harvard JOLT 2018, Is Privacy Privacy? 2018), Census 2020 DAS governance and reconstruction-attack record, DP auditing frameworks (arXiv:2210.08643, 2506.16666, 2509.08704, 2602.17454), and the sweep-digest reconstruction-bound corpus (Balle 2022, Kaissis 2023, Riess 2024, Biswas 2024/2025, Saeidian 2024, Pathiraja-Reiter 2026). This absence is reported per D2 rules and is NOT evidence of novelty — each conjunct is individually covered and two conjuncts co-occur in single works (Nissim 2018: (ii)+(a PIA-analog); Abowd-Schmutte/Census: (iii)-adjacent+(reconstruction-motivated Φ governance)).)

CTR-LR-02 — NARROWED (moderate-high, 75%)

Residual restatement (what now survives). Only the four-way JOINT composition survives: a PRICED $R(t)$ that is simultaneously (i) adversary-relative in the reconstruction-robustness sense — the object itself belongs to Balle-Cherubin-Hayes 2022 / Kaissis 2023 / Riess 2024 and must be cited as source; (ii) drifting via adversary background-information accumulation, explicitly distinguished from DP-composition depletion (Zhang-Zhu 2021), valuation/reward drift (Xu 2017, CHRONOS 2026, Bo mean-field 2025), and aging-protects direction (Age-Dependent DP 2022); (iii) held as structurally inalienable subject-owned durable capital, answering Cofone’s Calabresi-Melamed objections to propertization; and (iv) with the observer-to-subject gap decomposed as venue-exclusion/market-position rent shown NON-REDUCIBLE to Acemoglu et al.’s data-externality account and distinct in mechanism from Bergemann-Bonatti 2024’s platform-position rent. No per-leg or per-pair first-ness (drift+pricing, adversary-relative budget, drifting durable capital, inalienable compensation) may be claimed anywhere in the paper.

Basis. Five independent search scopes (~30 distinct queries) each found single-leg and pair coverage but no joint-quadruple hit; refuter fetched the closest pair-threat (arXiv:2101.04357) directly. Coverage of the legs is CONFIRMED by resolvable art (Zhang-Zhu, Balle, Farboodi-Veldkamp, Cofone, Acemoglu, Bergemann-Bonatti, Pathiraja-Reiter); absence of the joint composition is scope-limited, and two formal rival decompositions of the same gap (externality; platform rent) impose a real non-reducibility burden that keeps this below SURVIVES.

Covering art found by the refuter seat.

- [partial] Tao Zhang, Quanyan Zhu, “On the Differential Private Data Market: Endogenous Evolution, Dynamic Pricing, and Incentive Compatibility”, arXiv:2101.04357 (2021) (2021) — Prices privacy loss that accumulates across periods via DP composition, with dynamically set per-period epsilon and compensation under IC/IR constraints. Defeats the claim’s contrast ‘time-drifting vs static budget’ as a standalone novelty: a priced, time-evolving DP loss already exists. Still adversary-independent epsilon, transactional (alienable), no rent decomposition — quadruple untouched but the drift+pricing pair is anticipated. <https://arxiv.org/abs/2101.04357>
- [partial] Vincent Conitzer, Curtis R. Taylor, Liad Wagman, “Hide and Seek: Costly Consumer Privacy in a Market with Repeat Purchases”, Marketing Science 31(2):277-292 (2012) (2012) — Mainstream-economics precedent where the observer’s knowledge of the subject accumulates over time through repeat interactions and the subject pays a price (costly anonymity) to reset it — a priced, observer-knowledge-drifting privacy model. Covers the intuition behind adversary-side accumulation driving privacy value over time, without a reconstruction bound, capital holding, or rent decomposition. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2160895
- [partial] Lei Xu et al., “Dynamic Privacy Pricing: A Multi-Armed Bandit Approach With Time-Variant Rewards”, IEEE Trans. Information Forensics and Security (2017) (2017) — Data-market pricing of privacy where valuations are explicitly time-variant and prices adapt online — prior art for ‘dynamic privacy pricing’ as a phrase and mechanism; drift is on valuations/rewards, not on an adversary-relative reconstruction bound. https://www.researchgate.net/publication/316236814_Dynamic_Privacy_Pricing_A_Multi-Armed_Bandit_Approach_With_Time-Variant_Rewards
- [partial] Borja Balle, Giovanni Cherubin, Jamie Hayes, “Reconstructing Training Data with Informed Adversaries”, IEEE S&P 2022, arXiv:2201.04845 (2022) — Origin of the adversary-parameterized reconstruction-robustness (ReRo) bound: adversary-relative reconstruction bounds as a formal object are known art (extended by Kaissis et al. arXiv:2307.03928 and arXiv:2402.12861). Unpriced, static, no capital/rent — covers the ‘adversary-relative reconstruction bound’ leg alone. <https://arxiv.org/abs/2201.04845>
- [partial] Daron Acemoglu, Ali Makhdoumi, Azarakhsh Malekian, Asuman Ozdaglar, “Too Much Data: Prices and Inefficiencies in Data Markets”, AEJ: Microeconomics 14(4) (2022) (2022) — Formal competing decomposition of the depressed subject-side price via the data/correlation externality — an accumulating, other-subjects-relative information channel. The market-position-rent (venue-exclusion) decomposition must be shown non-reducible to this externality account. <https://www.aeaweb.org/articles?id=10.1257/mic.20200200>
- [partial] Dirk Bergemann, Alessandro Bonatti, “Data, Competition, and Digital Platforms”, American Economic Review 114(8):2553-2595 (2024) (2024) — Nearest formal precedent for locating the observer-to-subject value gap in the platform’s market position (control of match information and on/off-platform access) — the rent-decomposition leg is anticipated in spirit and must be differentiated in mechanism. <https://www.aeaweb.org/articles?id=10.1257/aer.20230478>
- [partial] Maryam Farboodi, Laura Veldkamp, “A Model of the Data Economy”, NBER WP 28427 / July 2024 version (2021-2024) — Data as a distinct, durable, dynamically depreciating capital stock with Bayes-law (information-theoretic) drift — covers the ‘time-drifting durable data-capital’ pair from the observer side; no privacy metric, no inalienability, no rent decomposition.

https://business.columbia.edu/sites/default/files-efs/citation_file_upload/DataGrowth_july2024.pdf

- [partial] Imanol Arrieta-Ibarra, Leonard Goff, Diego Jimenez-Hernandez, Jaron Lanier, E. Glen Weyl, “Should We Treat Data as Labor? Moving Beyond ‘Free’”, AEA Papers & Proceedings 108 (2018) (and Posner & Weyl, *Radical Markets*, 2018) (2018) — Prices a subject-attached, non-transferable-in-stock factor (data-as-labor flow with dividends to subjects) — prior art for compensating subjects while the underlying source remains attached to the person, compressing the ‘subject-owned, priced, non-alienated’ framing; no reconstruction bound, no adversary model, no drift mechanics. <https://www.aeaweb.org/conference/2018/preliminary/paper/2Y7N88na>
- [partial] Ignacio N. Cofone, “Beyond Data Ownership”, 43 *Cardozo Law Review* (2021) (2021) — Maps data rights onto the Calabresi-Melamed property/liability/inalienability grid and argues propertization fails on asymmetric information and inferred-data leakage — the structural-inalienability leg is occupied conceptually; the claim must show why ‘inalienable durable capital’ evades these objections (alongside Malgieri & Custers, “Pricing privacy – the right to know the value of your personal data”, *CLSR* 2018). https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3564480
- [partial] Meng Zhang, Ermin Wei, Randall Berry, Jianwei Huang, “Age-Dependent Differential Privacy”, *IEEE Trans. Information Theory* / arXiv:2209.01466 (2022) (2022) — A formally time-indexed privacy guarantee (privacy as an explicit function of data age), alongside the continual-observation strand (Dwork et al. 2010; Cao et al. temporal privacy leakage arXiv:1711.11436; temporally discounted DP arXiv:1908.03995) — ‘first time-indexed privacy quantity’ is unavailable as a novelty basis; drift direction and absence of pricing are the only distinguishers. <https://arxiv.org/abs/2209.01466>
- [partial] Ruwimal Y. Pathiraja, Jerome P. Reiter, “Setting the Privacy Budget in Differential Privacy by Bounding Adversaries’ Odds of Learning Sensitive Information”, arXiv:2607.04004 (2026) (2026) — Makes the operative privacy budget adversary-relative (calibrated to a specified adversary’s posterior odds) — the adversary-relativity leg of the budget-setting step is now occupied in the DP mainstream; static and unpriced. <https://arxiv.org/abs/2607.04004>
- [partial] Joydeep Chandra, “CHRONOS: Temporally-Aware Multi-Agent Coordination for Evolving Data Marketplaces”, arXiv:2605.23887 (2026) (2026) — Joins time-drifting valuation and privacy-budget consumption in one marketplace system — with Bo et al. (arXiv:2512.18296, 2512.21585) shows the DP-markets strand actively adding dynamics and strategic pricing; privacy object remains a DP budget, no inalienability or rent decomposition. <https://arxiv.org/abs/2605.23887>

Absence report (D2). `not_found_in_scope`(a single work jointly pricing an adversary-relative reconstruction bound that drifts over time via adversary background-information accumulation AND holding it as inalienable subject-owned durable capital AND decomposing the observer-to-subject value gap as market-position/venue-exclusion rent). Searched (8 distinct WebSearch queries beyond the sweep digest, plus targeted WebFetch of arXiv:2101.04357): privacy depreciation/decay pricing with adversary knowledge accumulation; data-as-labor/inalienable data rights pricing (Posner-Weyl lineage); dynamic anonymity pricing under repeat purchases (Conitzer-Taylor-Wagman lineage); adversary-dependent privacy pricing with time-varying risk and reconstruction bounds; information-rent/platform-intermediary decompositions of the personal-data value gap; inalienable-right and human-capital pricing analogies for personal data; continual-observation privacy-loss accumulation with compensation/budget-depletion pricing; and reconstruction-

bound-priced data valuation 2025-2026. Every hit covers at most one leg or one pair (drift+pricing: Zhang-Zhu 2021, Xu et al. 2017, CHRONOS 2026; adversary-relativity+reconstruction: Balle et al. 2022 / ReRo line; drift+durable-capital: Farboodi-Veldkamp; inalienability+compensation: data-as-labor, Cofone, Malgieri-Custers; gap decomposition: Bergemann-Bonatti 2024, Acemoglu et al. 2022). Absence of a joint-quadruple hit is a scope-limited search result, not proof of novelty.

CTR-LR-03 — NARROWED (moderate-high, 75%)

Residual restatement (what now survives). Survives only as the specific identification: the measured WTA/WTP floor and the IEEE 7012-2025 proposer inversion as TWO EVALUATIONS OF ONE preference-free subgame-perfect share function of proposer identity, with the subject share written as an explicit venue-exclusion parameter $s = h_X$. Stripped: any claim to first ultimatum-game treatment of privacy (Rajtmajer-Squicciarini 2017); the SPE-share-of-proposer mathematics (Stahl-Rubinstein, known art); and the structural-not-cognitive floor thesis AS SUCH, which is formally occupied by Gu 2024 (intermediary bargaining power) and Fallah et al. 2024 (competition benefits buyers not users) and qualitatively by Zuiderveen Borgesius 2017. Added burden: an explicit identification argument separating venue exclusion from the choice-architecture/reference-dependence rival (Lin & Strulov-Shlain 2023; arXiv:2607.27239; Fradkin), which explains large floors with no market-structure parameter. The 7012-as-permutation formalization itself remains unclaimed — no indexed work links MyTerms/7012 to any game-theoretic analysis, and no Stackelberg privacy-pricing variant makes the subject the leader.

Basis. Two refuter/scout scopes ran 9-12 targeted queries including direct 7012+game-theory and role-reversal Stackelberg searches with zero hits (a meaningful dog-that-didn't-bark given the standard published January 2026); but the structural-floor component is independently formalized (Gu, Fallah) and the empirical floor has a strong live cognitive rival, so the residue is genuinely smaller than stated. Strongest surviving residue of the five.

Covering art found by the refuter seat.

- [partial] Sarah Rajtmajer, Anna Squicciarini et al., “An Ultimatum Game Model for the Evolution of Privacy in Jointly Managed Content”, Decision and Game Theory for Security (GameSec), Springer LNCS (2017) — Prior art applying the ultimatum-game formalism inside the privacy domain (co-managed content negotiation). It does not touch WTA/WTP, venue exclusion, or proposer inversion, but it destroys any implicit claim to being the first ultimatum-game treatment of privacy; the residue must be stated as the specific WTA/WTP-plus-7012 identification, not as ‘ultimatum model of privacy’. https://link.springer.com/chapter/10.1007/978-3-319-68711-7_7
- [partial] Dale O. Stahl, “Subgame Perfection in Ultimatum Bargaining Trees” (Stahl–Rubinstein lineage: SPE shares as a function of the bargaining tree / who proposes) (c. 2007) — The formal engine of the claim — that the SPE share is a function of proposer identity / move order, and that permuting the proposer carries the responder from the floor to an interior share — is textbook Stahl/Rubinstein content, explicitly analyzed across bargaining-tree structures. The claim cannot own the share-function-of-proposer-identity mathematics; only the privacy-domain mapping is left. https://laits.utexas.edu/~stahl/experimental/ult_trees.pdf
- [partial] Jiadong Gu, “Data Trade and Consumer Privacy”, arXiv:2406.12457 (econ.TH) (2024 (v3 2026)) — Formal result that the data subject’s realized privacy/surplus is governed by the intermediary’s bargaining power, and that reducing that power protects the subject — a structural market-position determinant of the subject share, covering the ‘floor is structural

not cognitive' thesis in a formal model, though via screening rather than an SPE share function of proposer identity. <https://arxiv.org/abs/2406.12457>

- [partial] Alireza Fallah, Michael I. Jordan, Ali Makhdoumi, Azarakhsh Malekian, "On Three-Layer Data Markets", arXiv:2402.09697 (2024 (v4 2025)) — Equilibrium analysis in which the subject's low share is a consequence of market structure/position (competition benefits buyers, not users) — independent formal cover for the structural (non-preferential) account of the subject's floor, without ultimatum unification or a proposer-inversion remedy. <https://arxiv.org/abs/2402.09697>
- [partial] Frederik Zuiderveen Borgesius et al., "Tracking Walls, Take-It-Or-Leave-It Choices, the GDPR, and the ePrivacy Regulation", European Data Protection Law Review (re-hosted arXiv:2510.25339) (2017) — Explicit legal-economic treatment of privacy consent as a take-it-or-leave-it offer with no negotiation venue — the qualitative structural reading of the floor (consent not freely given because the subject cannot counter-offer), predating and paralleling the Richards–Hartzog lineage; no SPE model, no share function, no h_X . <https://arxiv.org/pdf/2510.25339>
- [partial] Lijun Bo & Weiqiang Chang, "Privacy Data Pricing: A Stackelberg Game Approach", arXiv:2512.18296 (2025) — Formalizes leader/proposer identity in a privacy-pricing game — closest formal object to proposer-identity parameterization; my role-reversal search confirmed no variant in this lineage makes the subject the leader, so the permutation leg remains open. <https://arxiv.org/abs/2512.18296>
- [partial] Tesary Lin & Avner Strulov-Shlain, "Choice Architecture, Privacy Valuations, and Selection Bias in Consumer Data" (2023 (rev. 2024)) — Strongest empirical statement of the rival cognitive/architectural account of the measured WTA/WTP floor; it covers the empirical-floor leg of the claim and forces the structural ($s = h_X$) reading to carry an explicit identification argument distinguishing venue exclusion from elicitation architecture. <https://arxiv.org/pdf/2308.13496>
- [partial] IEEE Standards Association, "IEEE 7012-2025 — Standard for Machine Readable Personal Privacy Terms" (published January 2026) (2026) — The proposer-inversion instrument exists as a ratified artifact self-described as making the individual first party; covers the remedy leg as engineering/advocacy fact, with no economic model of why the inversion moves the share — leaving only the formalization open. <https://ieeexplore.ieee.org/document/11360682>

Absence report (D2). `not_found_in_scope`(any published work casting the empirically measured privacy WTA/WTP floor and the IEEE 7012/MyTerms proposer inversion as two evaluations of ONE preference-free subgame-perfect-equilibrium share function of proposer identity, or writing the subject share as a structural venue-exclusion parameter $s = h_X$). Searched 2026-08-17, nine distinct web queries beyond the sweep digest: (1) ultimatum game + privacy WTA/WTP bargaining; (2) take-it-or-leave-it privacy consent + subgame perfect + data market subject share; (3) IEEE 7012/MyTerms + economics/game theory/bargaining-power/first-party; (4) privacy paradox WTA/WTP gap + structural bargaining-power (vs preference) explanations; (5) Rubinstein bargaining + personal data + outside option + proposer role; (6) data subjects as ultimatum responders / platform as proposer formal models; (7) ultimatum game analogies to privacy policies / adhesion contracts; (8) MyTerms/IEEE P7012 + ultimatum/bargaining/equilibrium (zero hits linking the standard to any game-theoretic analysis); (9) Stackelberg privacy pricing with user/consumer as leader (role reversal) — none found; the entire Stackelberg data-pricing lineage keeps the platform/collector as leader. Also searched missing-market/Coase structural accounts of the privacy

valuation gap and consent-as-ultimatum law-review framings. Absence is reported as absence in this scope, not as proof of novelty.

CTR-LR-04 — NARROWED (moderate, 68%)

Residual restatement (what now survives). The DESTINATION is occupied: ‘the redistributive object is position rent, not a wage’ is stated nearly verbatim in Mohades-Savona 2026 (marginal product attributed to data owners reflects monopoly position, via Marxian rent theory; early-stage draft — track to publication before relying on priority), and is convergently reached by Sadowski 2019, O’Reilly-Strauss-Mazzucato 2024 (attention rents), and Viljoen 2021, who runs the structurally isomorphic internal-defeat-then-respecify move. Acemoglu 2022 supplies a formal near-zero-marginal-price result and Vincent-Hecht-Sen 2019 ran the diminishing-returns substrate against data strikes but patched the programme. What survives: the specific DIRECTED falsifier route — Bajari et al. 2019 flat returns-to-scale deployed as an internal empirical defeat of the marginal-product wage logic, concluding self-defeat rather than patching — landing on venue-access/clearing-position rent INSIDE a reconstruction-bound / IEEE 7012 frame. The paper must position this route against three convergent-but-distinct falsifiers (Mohades-Savona: Marxian rent; Acemoglu: substitutability; Viljoen: relationality) and against the competing liability re-specification (Zhang-Zhu 2025).

Basis. Refuter did a full-text read of the strong candidate plus 9 targeted queries; the conclusion-space is demonstrably crowded (one strong + four partial candidates), leaving only the argument-route composition. Confidence is moderate rather than high because Mohades-Savona is an unpublished draft whose final form could expand to cover more of the route, and because route-versus-destination distinctions are the kind reviewers discount.

Covering art found by the refuter seat.

- [strong] Siavash Mohades & Maria Savona, “Data Rent and Surplus Value in the Digital Economy”, LUISS LEAP / Maastricht / Sussex working paper (early-stage draft, do not cite), January 19, 2026 (2026) — Directly occupies the claim’s re-specification leg: engages Posner-Weyl’s data-as-labor by name, formally distinguishes differential vs absolute DATA RENT (adapted from Marx’s ground rent) arising from monopolistic control of access, and states explicitly that “the ‘marginal product’ that neoclassical theory attributes to data owners reflects not their productive contribution but their monopoly position” and that “data rent is not a payment for data’s ‘marginal product’”. That is the claim’s conclusion — redistributive object = position rent, not a wage — published before WP-14. It reaches it via Marxian rent theory rather than the Bajari flat-returns falsifier, and its remedy is data sharing/commons rather than subject-side venue-access clearing position; no reconstruction bound or IEEE 7012. https://leap.luiss.it/wp-content/uploads/2026/02/mohades_paper.pdf
- [partial] Nicholas Vincent, Brent Hecht, Shilad Sen, “‘Data Strikes’: Evaluating the Effectiveness of a New Form of Collective Action Against Technology Companies”, The Web Conference (WWW) 2019 (2019) — The closest prior art to RUNNING the falsifier: an internal, empirical application of diminishing returns-to-data to the data-labour lever itself, simulating data strikes on recommender systems and showing small withdrawals have limited effect precisely because marginal records add little. Same year and same empirical substrate as Bajari 2019. It stops short of the claim’s conclusion — it concludes strikes need moderate scale (patching the programme) rather than declaring the wage remedy self-defeating and re-specifying the object as rent. Companion CDC paper (CSCW 2021, dl.acm.org/doi/10.1145/34449177) quan-

tifies the same diminishing-returns constraint. <https://dl.acm.org/doi/10.1145/3308558.313742>

- [partial] Salome Viljoen, “A Relational Theory of Data Governance”, Yale Law Journal 131(2) (2021); popular version “Data as Property?”, Phenomenal World (2020) (2021) — Structurally isomorphic falsifier-plus-re-specification: argues the propertarian/data-as-labor “market rate for personal data” remedy is mis-specified because data’s value is population-level and not reducible to individual contributions, and re-specifies the redistributive/governance object away from individual wages. Mechanism differs (relationality/sociality, not flat returns-to-scale evidence) and the landing differs (democratic collective governance, not venue-access rent), but the argumentative move — internal defeat of the wage frame forcing re-specification of the object — is prior art. <https://yalelawjournal.org/feature/a-relational-theory-of-data-governance>
- [partial] Daron Acemoglu, Ali Makhdoumi, Azarakhsh Malekian, Asu Ozdaglar, “Too Much Data: Prices and Inefficiencies in Data Markets”, American Economic Journal: Microeconomics 14(4): 218-256 (2022) (2022) — Formalizes why the price of an individual’s marginal record is structurally depressed toward zero (data externalities / substitutability across correlated users), implying no bargaining-toward-marginal-product remedy can deliver meaningful individual compensation — the economic core of the self-defeat, derived from substitutability rather than flat returns-to-scale, with remedies in market regulation/shutdown rather than venue-rent redistribution. <https://www.aeaweb.org/articles?id=10.1257/mic.20200200>
- [partial] Tim O’Reilly, Ilan Strauss, Mariana Mazzucato, “Algorithmic attention rents: A theory of digital platform market power”, Data & Policy 6 (2024) (2024) — Prior art for reading platform data value as RENT extracted from control of the allocation/matching venue (attention allocation position) rather than from productive contribution — the same clearing-position-rent object the claim re-specifies toward, without any engagement with the data-as-labour wage remedy or the flat-returns falsifier. <https://www.cambridge.org/core/journals/data-and-policy/article/algorithmic-attention-rents-a-theory-of-digital-platform-market-power/D85FE41F6CF99FC57DDFB2B2B63491C5>

Absence report (D2). not_found_in_scope(9 web searches + full-text read of Mohades-Savona 2026 across: [“data as labor” critique diminishing returns/marginal value wage self-defeating], [Bajari returns-to-scale + data-as-labor/data-dividend critique], [data dividend criticism marginal value near zero], [Acemoglu Too Much Data substitutability + data unions], [Viljoen relational theory vs propertarian compensation], [Sadowski data-as-capital rent], [Savona data rents redistribution vs wages], [Vincent-Hecht data strikes diminishing returns], [Posner-Weyl marginal product critique], [self-defeating data-dividend + rent-instead phrasing]): no work was found that (a) deploys Bajari et al. 2019 flat returns-to-scale evidence AS a directed falsifier concluding the data-as-labour wage remedy self-defeats, AND (b) re-specifies the redistributive object as venue-access/clearing-position rent in the same chain, AND (c) does either inside a reconstruction-bound or IEEE 7012 frame. Absence of this composition is reported as search-scope absence, not as proof of novelty.

CTR-LR-05 — NARROWED (moderate, 72%)

Residual restatement (what now survives). Stripped: ‘first to make auxiliary/background-information accumulation the shelf-life clock’ (Altman et al. 2018 do it in governance form; Rocher et al. 2019 quantitatively; Acemoglu 2022 economically via the correlation externality); ‘a durability/depreciation term for data-as-capital’ (Veldkamp 2023; Farboodi-Veldkamp; NBER w35266 endogenous depreciation; Coyle-Manley 2024; Valavi 2022); and ‘Mosca-style shelf-life reasoning in

economics venues’ (Mascelli-Rodden FEDS 2025-093; arXiv:2603.01091). Survives as the two-part composition: (a) writing a Mosca-FORM inequality whose clock is an information-theoretic reconstruction bound (Dinur-Nissim / QIF / Bayes-capacity sense, e.g. arXiv:2406.13569) in place of cryptographic break-time, AND (b) carrying that substituted clock into a SUBJECT-OWNED capital-asset valuation — with the clock explicitly distinguished from Veldkamp-style relevance-decay on both axes (adversary-side accumulation vs statistical obsolescence; subject-owned vs firm-owned asset).

Basis. Sixteen distinct queries across two scopes found the ingredient (QIF reconstruction bounds), the template (Mosca as portable practitioner framework), and the destination class (data-capital depreciation) separately but never the substitution or its valuation carriage; however Altman 2018 pre-empts the conceptual half and the practitioner generalization of Mosca weakens the inventiveness of re-templating, so only the formal composition survives. One flagged unresolved item (ResearchGate-only time-dependent HNDL framework) needs manual follow-up before submission.

Covering art found by the refuter seat.

- [strong] Laura Veldkamp, “Valuing Data as an Asset”, Review of Finance 27(5); Maryam Farboodi, Laura Veldkamp, “A Model of the Data Economy” (NBER WP 28427; July 2024 version) (2021-2024) — Covers the broader claim shape as the digest already flags: data as long-lived, depreciating capital with a formal, information-dynamics-driven depreciation rate (Bayes/Kalman precision decay). Firm-owned and obsolescence-clocked, not adversary-clocked or subject-owned — but it is the strongest covering art for ‘a durability term for data-as-capital’ as such. https://business.columbia.edu/sites/default/files-efs/citation_file_upload/DataGrowth_july2024.pdf
- [partial] Micah Altman, Alexandra Wood, David O’Brien, Salil Vadhan, Urs Gasser, “Practical approaches to big data privacy over time”, International Data Privacy Law 8(1) (2018) — Anticipates the conceptual half of the substitution: protection afforded by SDL/de-identification has a finite horizon set by accumulating auxiliary (background) information, and governance should be designed over the data lifetime. This is ‘background-information accumulation as the durability clock’ in governance form — it lacks the Mosca-form inequality and any capital-asset valuation, but it strips the claim of ‘first to make auxiliary-info accumulation the shelf-life clock’. <https://academic.oup.com/idpl/article/8/1/29/4930711>
- [partial] Luc Rocher, Julien M. Hendrickx, Yves-Alexandre de Montjoye, “Estimating the success of re-identifications in incomplete datasets using generative models”, Nature Communications 10:3069 (2019) — Formal, quantitative model of re-identification success showing anonymization is not future-proof as datasets accumulate — an operationalized, adversary-relative reconstruction-risk quantity that grows with available background data. Attack-side and governance-facing; no inequality, no valuation, but it is the standard citation a reviewer would raise against ‘a finite, adversary-relative, drifting reconstruction bound’ being new. <https://www.nature.com/articles/s41467-019-10933-3>
- [partial] Chao Li, Daniel Yang Li, Gerome Miklau, Dan Suciu, “A Theory of Pricing Private Data”, ICDT 2013 / arXiv:1208.5258 (with Ghosh-Roth 2011 auction lineage) (2012) — Covers the subject-owned valuation half in static form: data subjects are compensated in proportion to an information-theoretic privacy-loss quantity (differential-privacy leakage) — i.e., an information-theoretic bound already prices subject-owned data. No time clock and no Mosca-form inequality, but it blocks any reading of the claim as ‘first to price subject-owned data by an information-theoretic bound’. <https://arxiv.org/pdf/1208.5258>

- [partial] Justin Hsu, Marco Gaboardi, Andreas Haeberlen, Sanjeev Khanna, Arjun Narayan, Benjamin C. Pierce, Aaron Roth, “Differential Privacy: An Economic Method for Choosing Epsilon”, IEEE CSF 2014 / arXiv:1402.3329 (2014) — Bridges an information-theoretic privacy bound directly to economic compensation owed to data subjects for the risk they bear — the bound-to-subject-economics bridge, static in time. Reinforces that only the temporal/inequality composition, not the economics-of-the-bound, is residually open. <https://arxiv.org/pdf/1402.3329>
- [partial] Lin William Cong, Danxia Xie, Longtian Zhang, “Knowledge Accumulation, Privacy, and Growth in a Data Economy”, Management Science 68(9) / arXiv:2109.10028 (2022) — Consumer-generated (subject-side) data enters a capital/knowledge-accumulation growth model with an explicit privacy-infringement cost that evolves dynamically — subject-provided data as a dynamic factor with a time-path privacy cost. The privacy cost is a preference/externality term, not a reconstruction bound, and there is no shelf-life inequality; but it occupies the ‘subject-side data capital with dynamic privacy cost’ ground. <https://arxiv.org/abs/2109.10028>
- [partial] Daron Acemoglu, Ali Makhdoumi, Azarakhsh Malekian, Asuman Ozdaglar, “Too Much Data: Prices and Inefficiencies in Data Markets”, AEJ: Microeconomics 14(4) (2022) — The data externality — the informativeness of others’ accumulated data erodes an individual’s privacy and depresses the price of their data — is economically the same mechanism as ‘background-information accumulation degrades the subject’s asset’. Never cast as a shelf-life inequality or durability term, but it pre-occupies the economic content of adversary-side accumulation lowering subject-side data value. <https://www.aeaweb.org/articles?id=10.1257/mic.20200200>
- [partial] Meng Zhang, Ermin Wei, Randall Berry, Jianwei Huang, “Age-Dependent Differential Privacy”, IEEE Trans. Information Theory / arXiv:2209.01466 (2022) — A formally time-indexed information-theoretic privacy quantity (privacy risk as a function of data age) — the same mathematical object class as the substituted clock, applied to utility trade-offs rather than asset valuation and without a Mosca-form inequality. <https://arxiv.org/abs/2209.01466>
- [partial] Jillian Mascelli, Megan Rodden, “Harvest Now Decrypt Later: Examining Post-Quantum Cryptography and the Data Privacy Risks for Distributed Ledger Networks”, Federal Reserve FEDS WP 2025-093 (2025) — Closest published move of Mosca-style shelf-life reasoning into data-privacy economics (retrospective confidentiality horizon on recorded personal/financial data at an economics institution). Clock remains cryptographic break-time; no reconstruction-bound substitution, no subject-owned asset valuation. <https://www.federalreserve.gov/econres/feds/files/2025093pap.pdf>

Absence report (D2). not_found_in_scope(8 WebSearch sweeps, 2026-08-17: (1) ‘shelf life’ + reconstruction attack + auxiliary information + time-horizon inequality; (2) Mosca-inequality analog for de-identification/anonymization lifetime under auxiliary-information accumulation; (3) privacy half-life / re-identification value-decay economics; (4) Dinur-Nissim reconstruction bound + economic valuation/pricing of personal data over time; (5) ‘harvest now, re-identify/deanonymize later’ for anonymized personal data; (6) ‘privacy depreciation’ as an economic asset model driven by adversary information accumulation; (7) Mosca theorem cross-cited with differential privacy/reconstruction/re-identification; (8) subject-owned inalienable data-as-capital valuation with adversary-indexed durability. No source was found that (a) writes a Mosca-FORM inequality (protection-lifetime + migration/exposure time vs clock) whose clock is an information-theoretic reconstruction bound (Dinur-Nissim/QIF) rather than cryptographic break-time, NOR (b) carries such a substituted clock into a valuation formula for a subject-owned capital asset. Absence of

found art is reported as search outcome only, not as proof of novelty.)

Bibliography additions proposed by run 03

74 new resolvable citations surfaced this run and judged worth adding to the WP-14 bibliography. Provenance-gated per run-02 practice: A4 (citation-verifier) re-verifies every entry against a primary record before any tier-A use; provenance_confidence below is the runtime's own label, not A4's.

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